§5.2

(36) $\int \frac{1}{1+\cos 2x}\,dx = \int \frac{1}{1+2\cos^2 x-1}\,dx$

$\cos 2x = 2\cos^2 x - 1$

$= \int \frac{1}{2}\cdot\left(\frac{1}{\cos x}\right)^2 dx$

$= \frac{1}{2}\int \sec^2 x\,dx = \frac{1}{2}\tan x + C$

(69) $\int \tan^2 x\,dx = \int (\sec^2 x - 1)\,dx = \tan x - x + C$

$\tan^2 x = \sec^2 x - 1$

§5.3

(1)(a) $\int 2x(x^2+1)^{23}\,dx = \int u^{23}\,du = \frac{u^{24}}{24} + C$

$= \frac{(x^2+1)^{24}}{24} + C$

$u = x^2+1$

$du = 2x\,dx$

(b) $\int \cos^3 x \sin x\,dx = -\int u^3\,du = -\frac{u^4}{4} + C$

$= -\frac{\cos^4 x}{4} + C$

$u = \cos x$

$du = -\sin x\,dx$

(7) (a) $\int x^2\sqrt{1+x}\,dx = \int (u-1)^2\sqrt{u}\,du$

$u = 1+x$

or $x = u-1$

$du = dx$

$= \int (u^2-2u+1)\cdot u^{1/2}\,du = \int \left(u^{2+\frac{1}{2}} - 2u^{1+\frac{1}{2}} + u^{\frac{1}{2}}\right)du$

$= \frac{u^{2+1+\frac{1}{2}}}{2+1+\frac{1}{2}} - 2\frac{u^{1+\frac{1}{2}+1}}{1+\frac{1}{2}+1} + \frac{u^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C$

$= \frac{2}{7}u^{7/2} - 2\frac{2}{5}u^{5/2} + \frac{2}{3}u^{3/2} + C$

$= \frac{2}{7}(x+1)^{7/2} - \frac{4}{5}(x+1)^{5/2} + \frac{2}{3}(x+1)^{3/2} + C$

(b) $\int (\csc(\sin x))^2 \cos x\,dx = \int (\csc(u))^2\,du = -\cot u + C$

$= -\cot(\sin x) + C$

$u = \sin x$

$du = \cos x\,dx$

(10) (a) $\int \frac{\sin 3\theta}{1+\cos 3\theta}\,d\theta = -\frac{1}{3}\int \frac{1}{u}\,du = -\frac{1}{3}\ln|u| + C =$

$u = 1+\cos 3\theta$

$du = -3\sin 3\theta\,d\theta$

$= -\frac{1}{3}\ln|1+\cos 3\theta| + C$

(34) $\int \frac{e^x+e^{-x}}{e^x-e^{-x}}\,dx \overset{\uparrow}{=} \int \frac{du}{u} = \ln|u|+C$

$= \ln|e^x-e^{-x}|+C$

$u = e^x-e^{-x}$

$du = (e^x+e^{-x})\,dx$

(35) $\int \frac{e^x}{1+e^{2x}}\,dx \overset{\uparrow}{=} \int \frac{du}{1+u^2} = \tan^{-1}(u)+C$

$= \tan^{-1}(e^x)+C$

$u = e^x$

$du = e^x\,dx$

(59) $\int (\ln(e^x)+\ln(e^{-x}))\,dx = \int (x+(-x))\,dx = \int 0\,dx = C$

(61) (a) $\int \frac{dx}{\sqrt{9-x^2}} = \int \frac{1}{\sqrt{9}\sqrt{1-\left(\frac{x}{3}\right)^2}}\,dx = \frac{1}{3}\int \frac{1}{\sqrt{1-\left(\frac{x}{3}\right)^2}}\,dx$

$u = \frac{x}{3}$

$du = \frac{1}{3}dx$

$= \int \frac{1}{\sqrt{1-u^2}}\,du = \sin^{-1}(u)+C$

$= \sin^{-1}\left(\frac{x}{3}\right)+C$

(b) $\int \frac{dx}{5+x^2} = \frac{1}{5}\int \frac{1}{1+\left(\frac{x}{\sqrt{5}}\right)^2}\,dx =$ $u = \frac{x}{\sqrt{5}}$

$du = \frac{1}{\sqrt{5}}dx$

$= \frac{1}{\sqrt{5}}\int \frac{du}{1+u^2} = \frac{1}{\sqrt{5}}\tan^{-1}(u)+C$

$= \frac{1}{\sqrt{5}}\tan^{-1}\left(\frac{x}{\sqrt{5}}\right)+C$

(c) $$\int \frac{dx}{x\sqrt{x^2-n}} = \frac{1}{\sqrt{n}} \int \frac{dx}{x\sqrt{\left(\frac{x}{\sqrt{n}}\right)^2-1}} = \quad u = \frac{x}{\sqrt{n}},\ du = \frac{1}{\sqrt{n}}dx$$

$$= \frac{1}{\sqrt{n}} \sec^{-1}\left|\frac{x}{\sqrt{n}}\right| + C$$

(62)

(a) $$\int \frac{e^x}{4+e^{2x}}dx = \int \frac{du}{4+u^2} = \frac{1}{2}\int \frac{\frac{1}{2}du}{1+\left(\frac{u}{2}\right)^2} =$$

$u = e^x$, $du = e^x dx$

$$= \frac{1}{2}\tan^{-1}\left(\frac{u}{2}\right) + C$$

$$= \frac{1}{2}\tan^{-1}\left(\frac{e^x}{2}\right) + C$$

(b) $$\int \frac{dx}{\sqrt{9-4x^2}} = \frac{1}{2}\int \frac{du}{\sqrt{9-u^2}} = \frac{1}{2}\sin^{-1}\left(\frac{2x}{3}\right) + C$$

$u = 2x$, $du = 2dx$

(69) $$\frac{dy}{dx} = \sqrt{5x+1}, \quad y(3) = -2$$

$$y(x) \int \sqrt{5x+1}\,dx = \frac{1}{5}\int \sqrt{u}\,du = \frac{1}{5}\frac{u^{3/2}}{3/2} + C$$

$u = 5x+1$, $du = 5dx$

$$= \frac{2}{15}(5x+1)^{3/2} + C$$

$$-2 = y(3) = \frac{2}{15}(5\cdot 3+1)^{3/2} + C \rightsquigarrow C = -2 - \frac{2}{15}\cdot 64$$

$$\rightsquigarrow \boxed{y(x) = \frac{2}{15}(5x+1)^{3/2} - \frac{158}{15}} \quad \rightsquigarrow \boxed{C = \frac{-158}{15}}$$

Να υπολογιστούν τα ολοκληρώματα:

(i) $\int x\cos(3-x^2)\,dx = -\frac{1}{2}\int \cos u\,du = -\frac{1}{2}\sin u + c$

$u = 3-x^2$

$du = -2x\,dx$

$= -\frac{1}{2}\sin(3-x^2)+c$

(ii) $\int \sin x\cos x\,dx = \int u\,du = \frac{u^2}{2}+c = \frac{(\sin x)^2}{2}+c$

$u = \sin x$

$du = \cos x\,dx$

(iii) $\int \frac{3x^2}{\sqrt{1+x}}\,dx = \int \frac{3(u-1)^2}{\sqrt{u}}\,du = 3\int \frac{u^2-2u+1}{u^{1/2}}\,du$

$x = u-1$

$du = dx$

$= 3\int (u^{2-1/2} - 2u^{1-1/2} + u^{-1/2})\,du$

$= 3\int \left(u^{3/2} - 2u^{1/2} + u^{-1/2}\right)du$

$= 3\left(\frac{u^{5/2}}{5/2} - 2\frac{u^{3/2}}{3/2} + \frac{u^{1/2}}{1/2}\right)$

$= \frac{6}{5}(x+1)^{5/2} - \frac{4}{3}(x+1)^{3/2} + 6(x+1)^{1/2} + c$

(iv) $\int \sin x\cos 5x\,dx = \frac{1}{2}\int (\sin(x+5x) + \sin(x-5x))\,dx$

$\sin a\cos b = \frac{1}{2}(\sin(a+b)+\sin(a-b))$

$= \frac{1}{2}\int (\sin 6x + \sin(-4x))\,dx = \frac{1}{2}\int (\sin 6x - \sin 4x)\,dx$

$= \frac{1}{2}\left(-\frac{1}{6}\cos 6x\right) - \frac{1}{2}\left(\frac{1}{4}\cos 4x\right) + c$

(v) $\int \cos 3x \cos 5x \, dx = \frac{1}{2}\int (\cos(3x+5x) + \cos(3x-5x))$

$\cos a \cos b = \frac{1}{2}(\cos(a+b) + \cos(a-b))$

$= \frac{1}{2}\int (\cos 8x + \cos(-2x))dx = \frac{1}{2}\cdot\frac{1}{8}\sin 8x + \frac{1}{2}\cdot\frac{1}{2}\sin$ $+C$

(vi) $\int \sin^4 x \cos^3 x \, dx = \int \sin^4 x \cdot \cos^2 x \cos x \, dx$

$u = \sin x$, $du = \cos x \, dx$

$= \int \sin^4 x (1-\sin^2 x)\cos x \, dx$

$= \int u^4 (1-u^2)\, du =$

$= \int (u^4 - u^6)\, du =$

$= \frac{u^5}{5} - \frac{u^7}{7} + C =$

$= \frac{\sin^5 x}{5} - \frac{\sin^7 x}{7} + C$

(vii) $\int \sin^3 x \cos^6 x \, dx = \int \sin^2 x \sin x \cos^6 x \, dx$

$u = \cos x$, $du = -\sin x dx$

$= \int (1-\cos^2 x)\sin x \cos^6 x \, dx$

$= \int (1-u^2)(-1)u^6 \, du$

$= \int (-u^6 + u^8)\, du =$

$= \frac{u^9}{9} - \frac{u^7}{7} + C$

$= \frac{\cos^9 x}{9} - \frac{\cos^7 x}{7} + C$

(iii) $\int \frac{dx}{1+e^x} \overset{u=e^x}{=} \int \frac{1}{u(u+1)}\,du =$

$u = e^x$

$du = e^x dx$

$\frac{1}{u(u+1)} = \frac{A}{u} + \frac{B}{u+1}$

$1 = A(u+1) + Bu$

$u = -1 \quad 1 = -B \Rightarrow \boxed{B=-1}$

$u = 0 \quad 1 = A \Rightarrow \boxed{A=1}$

$= \int \left(\frac{1}{u} - \frac{1}{u+1}\right) du = \ln|u| - \ln|u+1| + C$

$= \ln\left|\frac{u}{u+1}\right| + C = \ln\left|\frac{e^x}{e^x+1}\right| + C$

(iv) $\int \frac{\cos x}{\sqrt{2-\sin^2 x}}\,dx \overset{u=\sin x}{=} \int \frac{du}{\sqrt{2-u^2}} = \frac{1}{\sqrt{2}} \int \frac{du}{\sqrt{1-\left(\frac{u}{\sqrt{2}}\right)^2}}$

$u = \sin x$

$du = \cos x\,dx$

$\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\frac{u}{\sqrt{2}} = \sin\theta$

$du = \sqrt{2}\cos\theta\,d\theta$

$\rightarrow = \frac{1}{\sqrt{2}} \int \frac{\sqrt{2}\cos\theta\,d\theta}{\sqrt{1-\sin^2\theta}} = \int \frac{\cos\theta}{\cos\theta}\,d\theta$

$= \theta + C = \sin^{-1}\left(\frac{u}{\sqrt{2}}\right) + C$

$= \sin^{-1}\left(\frac{\sin x}{\sqrt{2}}\right) + C$

(x) $\int x \tan^{-1}x \, dx = \int \left(\frac{x^2}{2}\right)' \tan^{-1}x \, dx$

$= \frac{x^2}{2}\tan^{-1}x - \int \frac{x^2}{2}\cdot\frac{1}{1+x^2}\,dx$

$= \frac{x^2}{2}\tan^{-1}x - \frac{1}{2}\int \frac{(x^2+1)-1}{x^2+1}\,dx$

$= \frac{x^2}{2}\tan^{-1}x - \frac{1}{2}\int\left(1-\frac{1}{x^2+1}\right)dx$

$= \frac{x^2}{2}\tan^{-1}x - \frac{x}{2} + \frac{1}{2}\tan^{-1}x + c$

(xi) $\int \cos(\ln x)\,dx$ Ekanpke $u = \ln x$

$du = \frac{1}{x}dx$

$\Rightarrow dx = x\,du$

$= e^u du$

$= \int (\cos u)\, e^u \, du$

$I = \int e^u \cdot \cos u \, du = \int e^u (\sin u)' du = e^u \cdot \sin u - \int e^u \sin u \, d$

$= e^u \sin u - \int e^u (-\cos u)' du$

$= e^u \sin u - \left(-e^u \cos u - \int e^u (-\cos u) du\right) =$

$= e^u \sin u + e^u \cos u - \int e^u \cos u \, du$

$= e^u \sin u + e^u \cos u - I$

Dnie

$2I = e^u(\sin u + \cos u) \rightsquigarrow \boxed{I = \frac{1}{2}e^u(\sin u + \cos u)}$